Diag Solutions

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1. **1.** Two concentric circles are of radii 5 cm and 3 cm. The length of the chord of the larger circle which touches the smaller circle is:

   Answer: (B)

   Solution:

   1. Let O be the center, AB be the chord of the larger circle touching the smaller circle at M.
   2. In $\triangle OMA$, $\angle OMA = 90^\circ$, $OA = 5$ cm, $OM = 3$ cm.
   3. By Pythagoras: $AM^2 = OA^2 - OM^2 = 25 - 9 = 16$. So $AM = 4$ cm.
   4. The perpendicular from the center bisects the chord, so $AB = 2 \times AM = 8$ cm.

   Derivation:

   $$AB = 2\sqrt{5^2 - 3^2} = 2\sqrt{16} = 8$$

   Circles — Concentric circles

2. **2.** If TP and TQ are two tangents to a circle with center O so that $\angle POQ = 110^\circ$, then $\angle PTQ$ is equal to:

   Answer: (C)

   Solution:

   1. In the quadrilateral OPTQ, the sum of angles is 360°.
   2. Tangents are perpendicular to radii at contact points, so $\angle OPT = 90^\circ$ and $\angle OQT = 90^\circ$.
   3. $\angle PTQ + \angle POQ + \angle OPT + \angle OQT = 360^\circ$.
   4. $\angle PTQ + 110^\circ + 90^\circ + 90^\circ = 360^\circ \implies \angle PTQ = 360^\circ - 290^\circ = 70^\circ$.

   Derivation:

   $$\angle PTQ = 180^\circ - 110^\circ = 70^\circ$$

   Circles — Tangents from an external point

3. **3.** Assertion (A): The length of a tangent drawn from an external point to a circle is always equal to the radius of the circle. Reason (R): Tangents drawn from an external point to a circle are equal in length.

   Answer: (D)

   Solution:

   1. The length of the tangent from an external point is not necessarily equal to the radius.

2. The theorem states that tangents drawn from an external point to a circle are equal in length.
3. Therefore, Assertion is false and Reason is true.

Derivation:

$$A : \text{False}, R : \text{True}$$

Circles — Tangents to a circle

4. **4.** Assertion (A): If the angle between two tangents drawn from a point P to a circle of radius r and center O is $60°$, then $OP = 2r$. Reason (R): The line segment joining the center to the external point bisects the angle between the two tangents.

Answer: (A)

Solution:

1. In $\triangle OAP$ (where A is point of contact), $\angle OPA = 60°/2 = 30°$.
2. $\sin(30°) = r/OP$, so $1/2 = r/OP$, which implies $OP = 2r$.
3. The reason explains the bisection property used in the calculation.

Derivation:

$$\sin(30°) = \frac{r}{OP}$$
$$\implies OP = 2r$$

Circles — Tangents to a circle

5. **5.** Assertion (A): A tangent to a circle is perpendicular to the radius at the point of contact. Reason (R): The perpendicular distance from the center of the circle to the tangent is equal to the radius of the circle.

Answer: (A)

Solution:

1. The theorem of tangents states that the tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. The shortest distance from the center to the line (tangent) is the radius, which confirms the perpendicularity.
3. Both statements are true and R explains A.

Derivation:

$$\text{Radius} \perp \text{Tangent}$$

Circles — Tangents to a circle

6. **6.** Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A tangent to a circle intersects the circle at only one point.

Answer: (A)

Solution:

1. A point inside a circle cannot have a line touching the circle at only one point.
2. A tangent is defined as a line intersecting the circle at exactly one point.
3. Both are true and R explains why no tangents can exist from an internal point.

Derivation:

$$\text{Internal point} \implies 0 \text{ tangents}$$

Circles — Tangents to a circle

7. **7.** Assertion (A): If the distance between two parallel tangents of a circle of radius 3 cm is 6 cm. Reason (R): Parallel tangents touch the circle at the ends of the diameter.

Answer: (A)

Solution:

1. The distance between two parallel tangents is equal to the diameter of the circle.
2. Diameter = 2× radius = $2 \times 3 = 6$ cm.
3. Both are true and R explains why the distance is the diameter.

Derivation:

$$d = 2r = 2(3) = 6$$

Circles — Tangents to a circle

8. **8.** If a tangent PA and PB from a point P to a circle with center O are inclined to each other at an angle of 80°, then find $\angle POA$.

Answer:

$$50°$$

Solution:

1. In quadrilateral OAPB, $\angle OAP = \angle OBP = 90°$ because the radius is perpendicular to the tangent at the point of contact.
2. In $\triangle OAP$ and $\triangle OBP$, OP bisects $\angle APB$, so $\angle APO = 40°$. In $\triangle OAP$, $\angle POA = 180° - (90° + 40°) = 50°$.

Derivation:

$$\angle APB = 80°$$
$$\implies \angle APO = 40°. \text{ In } \triangle OAP, \angle OAP = 90°. \angle POA = 180° - 90° - 40° = 50°.$$

Circles — Tangents to a circle

9. **9.** A quadrilateral ABCD is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, find the length of AD.

Answer:

$$3 \text{ cm}$$

Solution:

1. For a quadrilateral circumscribing a circle, the sums of opposite sides are equal: $AB + CD = BC + AD$.
2. Substitute the given values: $6 + 4 = 7 + AD$. Solving for AD gives $10 - 7 = 3$.

Derivation:

$$\begin{aligned} AB + CD &= BC + AD \\ \implies 6 + 4 &= 7 + AD \\ \implies AD &= 10 - 7 = 3 \text{ cm.} \end{aligned}$$

Circles — Properties of tangents

10. **10.** From a point Q, the length of the tangent to a circle is 24 cm and the distance of Q from the center is 25 cm. Find the radius of the circle.

Answer:

$$7 \text{ cm}$$

Solution:

1. Let O be the center and A be the point of contact. $\triangle OAQ$ is a right-angled triangle with $\angle OAQ = 90°$.
2. By Pythagoras theorem, $OQ^2 = OA^2 + AQ^2$. Substitute $25^2 = r^2 + 24^2$ and solve for r.

Derivation:

$$\begin{aligned} 25^2 &= r^2 + 24^2 \\ \implies 625 &= r^2 + 576 \\ \implies r^2 &= 49 \\ \implies r &= 7 \text{ cm.} \end{aligned}$$

Circles — Tangents to a circle

11. **11.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8 \text{ cm}$$

Solution:

1. Let AB be the chord of the larger circle touching the smaller circle at M. $OM \perp AB$ and $OM = 3$ cm, $OA = 5$ cm.
2. In right $\triangle OMA$, $AM = \sqrt{OA^2 - OM^2} = \sqrt{25 - 9} = 4$. Since M is the midpoint, $AB = 2 \times 4 = 8$.

Derivation:

$$AM = \sqrt{5^2 - 3^2} = \sqrt{16} = 4$$
$$\implies AB = 2 \times 4 = 8 \text{ cm}.$$

Circles — Tangents to a circle

12. **12.** If the angle between two tangents drawn from an external point P to a circle of radius r and center O is $60°$, find the length of OP.

Answer:

$$2r$$

Solution:

1. Let the tangents be PA and PB. OP bisects $\angle APB$, so $\angle APO = 30°$.
2. In $\triangle OAP$, $\sin(30°) = \frac{OA}{OP} = \frac{r}{OP}$. Since $\sin(30°) = 1/2$, we get $OP = 2r$.

Derivation:

$$\sin(30°) = \frac{r}{OP}$$
$$\implies \frac{1}{2} = \frac{r}{OP}$$
$$\implies OP = 2r.$$

Circles — Tangents to a circle

13. **13.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8 \text{ cm}$$

Solution:

1. Let O be the center, AB be the chord of the larger circle tangent to the smaller circle at point P.
2. OP is the radius of the smaller circle, so OP = 3 cm and OP ⊥ AB.
3. OA is the radius of the larger circle, so OA = 5 cm.
4. In right triangle OPA, $AP^2 = OA^2 - OP^2 = 5^2 - 3^2 = 16$, so AP = $4cm$.
5. Since the perpendicular from the center bisects the chord, AB = 2 * AP = 8 cm.

Derivation:

$$AP = \sqrt{OA^2 - OP^2} = \sqrt{25 - 9} = \sqrt{16} = 4 \text{ cm}; AB = 2 \times 4 = 8 \text{ cm}$$

Circles — Tangent properties

14. **14.** A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB+CD = AD + BC$.

Answer:

$$AB + CD = AD + BC$$

Solution:

1. Use the property that lengths of tangents drawn from an external point to a circle are equal.
2. Let the circle touch sides AB, BC, CD, DA at P, Q, R, S respectively.
3. $AP = AS, BP = BQ, CR = CQ, DR = DS$.
4. Summing the left and right sides of these equations leads to the result.

Derivation:

$$AB + CD = (AP + PB) + (CR + RD) = AS + BQ + CQ + DS = (AS + DS) + (BQ + CQ) = AD + BC$$

Circles — Tangents to a circle

15. **15.** If a tangent PA and PB from a point P to a circle with centre O are inclined to each other at an angle of $80°$, find $\angle POA$.

Answer:

$$50°$$

Solution:

1. In quadrilateral OAPB, $\angle OAP = \angle OBP = 90°$ (tangent is perpendicular to radius).
2. Sum of angles in quadrilateral is $360°$, so $\angle AOB = 180° - 80° = 100°$.
3. In $\triangle OAP$ and $\triangle OBP$, by RHS congruence, $\angle POA = \angle POB$.
4. Therefore, $\angle POA = \frac{1}{2}\angle AOB = 50°$.

Derivation:

$$\angle AOB = 180° - 80° = 100°; \angle POA = \frac{100°}{2} = 50°$$

Circles — Tangent properties

16. **16.** Prove that the parallelogram circumscribing a circle is a rhombus.

Answer:

$$AB = BC = CD = DA$$

Solution:

1. Since it is a parallelogram, opposite sides are equal: AB = CD and AD = BC.
2. From the property of circumscribed quadrilaterals, AB + CD = AD + BC.
3. Substitute CD with AB and BC with AD: 2AB = 2AD, hence AB = AD.
4. Since adjacent sides are equal, the parallelogram is a rhombus.

Derivation:

$$\begin{aligned} AB + CD &= AD + BC \\ \implies 2AB &= 2AD \\ \implies AB &= AD \end{aligned}$$

Circles — Tangents to a circle

17. **17.** A point P is 13 cm from the centre of a circle. The length of the tangent drawn from P to the circle is 12 cm. Find the radius of the circle.

Answer:

$$5 \text{ cm}$$

Solution:

1. Let O be the center and A be the point of contact on the circle.
2. Triangle OAP is a right-angled triangle at A because the radius is perpendicular to the tangent.
3. Hypotenuse OP = $13cm,$ tangent AP = $12cm.$
4. Using Pythagoras theorem, $OA^2 = OP^2 - AP^2 = 13^2 - 12^2 = 169 - 144 = 25$.
5. $OA = 5cm.$

Derivation:

$$OA = \sqrt{13^2 - 12^2} = \sqrt{169 - 144} = \sqrt{25} = 5 \text{ cm}$$

Circles — Tangents to a circle

18. **18.** Prove that the lengths of tangents drawn from an external point to a circle are equal.

Answer:

$$PA = PB$$

Solution:

1. Draw a circle with center O and an external point P.
2. Draw tangents PA and PB touching the circle at A and B respectively.
3. Join OA, OB, and OP.
4. Consider triangles OAP and OBP.

5. Use the property that the tangent at any point of a circle is perpendicular to the radius through the point of contact.
6. Use RHS congruence criterion to prove triangles are congruent.
7. Conclude that corresponding parts of congruent triangles are equal.

Derivation: In △ OAP and △ OBP : \\ 1. ∠ OAP = ∠ OBP = 90ˆ∘(Tangent ⊥ radius) \\ 2. OP = OP (Common side) \\ 3. OA = OB (Radii of the same circle) \\ By RHS congruence rule, △ OAP ≅ △ OBP . \\ Therefore, PA = PB (CPCT). *Circles — Tangents to a circle*

19. **19.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8cm$$

Solution:

1. Let O be the center of both circles.
2. Let AB be the chord of the larger circle touching the smaller circle at point P.
3. OP is the radius of the smaller circle, so $OP = 3$ cm.
4. OA is the radius of the larger circle, so $OA = 5$ cm.
5. Since AB is a tangent to the inner circle, $OP \perp AB$.
6. In right $\triangle OPA$, use Pythagoras theorem to find AP.
7. Since the perpendicular from the center bisects the chord, $AB = 2 \times AP$.

Derivation:

$$\begin{aligned} In \triangle OPA : OA^2 &= OP^2 + AP^2 \\ 5^2 &= 3^2 + AP^2 \\ 25 &= 9 + AP^2 \\ AP^2 &= 16 \\ \Rightarrow AP &= 4 \text{ cm.} \\ \text{Since OP} \perp AB, AP &= PB = 4 \text{ cm.} \\ AB &= AP + PB = 4 + 4 = 8 \text{ cm.} \end{aligned}$$

Circles — Properties of tangents

20. **20.** A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB + CD = AD + BC$.

Answer:

$$AB + CD = AD + BC$$

Solution:

1. Identify the points of contact on sides AB, BC, CD, and DA as P, Q, R, and S respectively.
2. Use the theorem that lengths of tangents from an external point to a circle are equal.
3. Express segments AP, BP, BQ, CQ, CR, DR, DS, and AS in terms of equality.
4. Sum the equations formed by these segments.
5. Rearrange the terms to group segments forming the sides of the quadrilateral.

Derivation: AP = AS (1), BP = BQ (2), CR = CQ (3), DR = DS (4) \\ Add (1), (2), (3), and (4): \\ (AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ) \\ AB + CD = AD + BC. *Circles — Tangents to a circle*

21. **21.** If a hexagon ABCDEF circumscribes a circle, prove that $AB + CD + EF = BC + DE + FA$.

Answer:

$$AB + CD + EF = BC + DE + FA$$

Solution:

1. Let the points of contact be P, Q, R, S, T, and U on sides AB, BC, CD, DE, EF, and FA respectively.
2. By the theorem of tangents from an external point, $AP = AF$, $BP = BQ$, $CQ = CR$, $DR = DS$, $ES = ET$, $FT = FU$ is incorrect; rather, label them appropriately.
3. Correct logic: $AP = AU, BP = BQ, CQ = CR, DR = DS, ES = ET, FT = FU$.
4. Sum the left side $AB + CD + EF$ using these relations.
5. Observe that the sum equals the sum of the remaining sides $BC + DE + FA$.

Derivation: Let points of contact be P_1, P_2, P_3, P_4, P_5, P_6 . \\ AB+CD+EF = (AP_1+P_1B) + (CP_3+P_3D) + (EP_5+P_5F) = (AP_6+P_2B) + (CP_2+P_4D) + (EP_4+P_6F) \\ Rearranging: (P_2B+CP_2) + (P_4D+EP_4) + (P_6F+AP_6) = BC + DE + FA . *Circles — Tangents to a circle*

22. **22.** A tower stands vertically on the ground. From a point on the ground, which is 15 m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60°. The height of the tower is:

Answer: (C)

Solution:

1. Let the height of the tower be h.
2. In the right triangle, $\tan(60°) = \frac{\text{height}}{\text{base}}$.
3. Substitute the values: $\sqrt{3} = \frac{h}{15}$.
4. Solve for h.

Derivation:

$$\tan(60^\circ) = \frac{h}{15}$$
$$\implies \sqrt{3} = \frac{h}{15}$$
$$\implies h = 15\sqrt{3} \text{ m}$$

Some Applications of Trigonometry — Heights and Distances

23. **23.** A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 30°. The length of the string is:

Answer: (D)

Solution:

1. Let L be the length of the string.
2. In the right triangle, $\sin(30^\circ) = \frac{\text{height}}{\text{hypotenuse}}$.
3. Substitute the values: $\frac{1}{2} = \frac{60}{L}$.
4. Solve for L.

Derivation:

$$\sin(30^\circ) = \frac{60}{L}$$
$$\implies \frac{1}{2} = \frac{60}{L}$$
$$\implies L = 120 \text{ m}$$

Some Applications of Trigonometry — Heights and Distances

24. **24.** The angle of elevation of the top of a pole from a point on the ground 20 m away from the foot of the pole is 45°. The height of the pole is:

Answer: (B)

Solution:

1. Let h be the height of the pole.
2. Use the relation $\tan(45^\circ) = \frac{h}{20}$.
3. Since $\tan(45^\circ) = 1$, we have $1 = \frac{h}{20}$.

Derivation:

$$\tan(45^\circ) = \frac{h}{20}$$
$$\implies 1 = \frac{h}{20}$$
$$\implies h = 20 \text{ m}$$

Some Applications of Trigonometry — Heights and Distances

25. 25. If the length of the shadow of a tower is $\sqrt{3}$ times the height of the tower, then the angle of elevation of the sun is:

Answer: (A)

Solution:

1. Let the height of the tower be h. Then the shadow length is $h\sqrt{3}$.
2. Let θ be the angle of elevation.
3. Use $\tan(\theta) = \frac{h}{h\sqrt{3}}$.
4. Simplify to find θ.

Derivation:

$$\begin{aligned} \tan(\theta) &= \frac{h}{h\sqrt{3}} = \frac{1}{\sqrt{3}} \\ &\implies \theta = 30^\circ \end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

26. 26. An observer 1.5 m tall is 28.5 m away from a chimney. The angle of elevation of the top of the chimney from her eyes is 45°. The height of the chimney is:

Answer: (C)

Solution:

1. Height of observer = 1.5 m. Distance = 28.5 m.
2. Height of chimney above observer's eye level $= 28.5 \times \tan(45^\circ) = 28.5$ m.
3. Total height $= 28.5 + 1.5$.

Derivation:

$$H = 28.5 + 1.5 = 30 \text{ m}$$

Some Applications of Trigonometry — Heights and Distances

27. 27. Assertion (A): The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower of height $30\sqrt{3}$ m, is 60°. Reason (R): In a right-angled triangle, $\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$.

Answer: (A)

Solution:

1. Identify the height (opposite) $= 30\sqrt{3}$ and base (adjacent) $= 30$.
2. Use the formula $\tan(\theta) = \frac{30\sqrt{3}}{30} = \sqrt{3}$.
3. Since $\tan(60^\circ) = \sqrt{3}$, the angle is 60°.

Derivation:

$$\begin{aligned} \tan(\theta) &= \frac{30\sqrt{3}}{30} = \sqrt{3} \\ &\implies \theta = 60^\circ \end{aligned}$$

Some Applications of Trigonometry — Angle of Elevation

28. 28. Assertion (A): If the length of the shadow of a vertical pole is equal to its height, then the angle of elevation of the Sun is 45°. Reason (R): The angle of elevation is defined only when the object is taller than the observer.

Answer: (C)

Solution:

1. Let height $= h$ and shadow $= h$.
2. $\tan(\theta) = \frac{h}{h} = 1$.
3. $\theta = 45°$. Assertion is true.
4. Reason is false because angle of elevation is defined by the line of sight relative to the horizontal, regardless of relative height.

Derivation:

$$\tan(\theta) = \frac{\text{height}}{\text{shadow}} = \frac{h}{h} = 1$$
$$\implies \theta = 45°$$

Some Applications of Trigonometry — Angle of Elevation

29. 29. Assertion (A): The angle of depression of an object from a point h meters above the ground is always equal to the angle of elevation of that point from the object on the ground. Reason (R): The angle of elevation and depression are alternate interior angles for parallel horizontal lines.

Answer: (A)

Solution:

1. By the properties of parallel lines, the line of sight creates alternate interior angles.
2. Thus, angle of elevation = angle of depression.

Derivation:

$$\angle\text{elevation} = \angle\text{depression (Alternate Interior Angles)}$$

Some Applications of Trigonometry — Angle of Depression

30. 30. Assertion (A): If a ladder of length 10 m reaches a wall at a height of 5 m, the angle made by the ladder with the ground is 30°. Reason (R): $\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}}$.

Answer: (A)

Solution:

1. Opposite = 5, Hypotenuse = 10.
2. $\sin(\theta) = \frac{5}{10} = 0.5$.
3. $\theta = 30°$.

Derivation:

$$\sin(\theta) = \frac{5}{10} = \frac{1}{2}$$
$$\implies \theta = 30^\circ$$

Some Applications of Trigonometry — Trigonometric Ratios

31. **31.** Assertion (A): As an observer moves closer to a tower, the angle of elevation of its top increases. Reason (R): $\tan(\theta)$ increases as θ increases from 0° to 90°.

Answer: (A)

Solution:

1. In $\tan(\theta) = \frac{h}{x}$, as x (distance) decreases, $\tan(\theta)$ increases.
2. As $\tan(\theta)$ increases, θ must increase.

Derivation:

$$\text{As } x \to 0, \tan(\theta) = \frac{h}{x} \to \infty, \text{ so } \theta \to 90^\circ$$

Some Applications of Trigonometry — Angle of Elevation

32. **32.** A ladder 15 m long makes an angle of 60° with the wall. Find the height of the point where the ladder touches the wall.

Answer:

$$7.5m$$

Solution:

1. Identify the triangle formed by the ladder (hypotenuse), the wall (adjacent side to 60 degrees), and the ground.
2. Use the trigonometric ratio $\cos(60^\circ) = \frac{\text{adjacent}}{\text{hypotenuse}}$ to find the height.

Derivation:

$$\cos(60^\circ) = \frac{h}{15}$$
$$\implies \frac{1}{2} = \frac{h}{15}$$
$$\implies h = 7.5 \text{ m}$$

Some Applications of Trigonometry — Trigonometric Ratios

33. **33.** The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 30°. Find the height of the tower.

Answer:

$$10\sqrt{3}m$$

Solution:

1. Use the tangent ratio $\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}}$ where $\theta = 30°$.
2. Substitute the given values and solve for the height h.

Derivation:

$$\tan(30°) = \frac{h}{30}$$
$$\implies \frac{1}{\sqrt{3}} = \frac{h}{30}$$
$$\implies h = \frac{30}{\sqrt{3}} = 10\sqrt{3} \text{ m}$$

Some Applications of Trigonometry — Angle of Elevation

34. **34.** A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60°. Find the length of the string.

Answer:

$$40\sqrt{3}m$$

Solution:

1. Represent the situation as a right triangle where the height is the opposite side and string length is the hypotenuse.
2. Use $\sin(60°) = \frac{\text{height}}{\text{length}}$ to solve for the length.

Derivation:

$$\sin(60°) = \frac{60}{L}$$
$$\implies \frac{\sqrt{3}}{2} = \frac{60}{L}$$
$$\implies L = \frac{120}{\sqrt{3}} = 40\sqrt{3} \text{ m}$$

Some Applications of Trigonometry — Real-world problems

35. **35.** A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.

Answer:

$$19\sqrt{3}m$$

Solution:

1. Determine the effective height of the building relative to the boy's eye level: $30 - 1.5 = 28.5$ m.

2. Use $\cot(30°)$ and $\cot(60°)$ to find the difference in ground distances.

Derivation:

$$d = 28.5(\cot(30°) - \cot(60°)) = 28.5(\sqrt{3} - \frac{1}{\sqrt{3}}) = 28.5(\frac{2}{\sqrt{3}}) = \frac{57}{\sqrt{3}} = 19\sqrt{3} \text{ m}$$

Some Applications of Trigonometry — Angle of Elevation

36. 36. From a point on the ground, the angles of elevation of the bottom and top of a transmission tower fixed at the top of a 20 m high building are 45° and 60° respectively. Find the height of the tower.

Answer:

$$20(\sqrt{3} - 1)m$$

Solution:

1. Let h be the height of the tower. The building height is 20.
2. Use $\tan(45°)$ to find the distance from the point to the building, then use $\tan(60°)$ for the total height.

Derivation:

$$\text{Distance } x = \frac{20}{\tan(45°)} = 20. \text{ Total height } (20 + h) = x\tan(60°) = 20\sqrt{3}.$$
$$\implies h = 20\sqrt{3} - 20 = 20(\sqrt{3} - 1) \text{ m}$$

Some Applications of Trigonometry — Multiple Angles of Elevation

37. 37. A tower stands vertically on the ground. From a point on the ground, which is 15 m away from the foot of the tower, the angle of elevation of the top of the tower is found to be 60°. Find the height of the tower.

Answer:

$$15\sqrt{3} \text{ m}$$

Solution:

1. Let h be the height of the tower and $d = 15$ m be the distance from the foot.
2. Use the tangent ratio: $\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$.
3. Substitute the values: $\tan(60°) = \frac{h}{15}$.
4. Solve for h: $h = 15 \times \sqrt{3}$.

Derivation:

$$\tan(60°) = \frac{h}{15}$$
$$\implies \sqrt{3} = \frac{h}{15}$$
$$\implies h = 15\sqrt{3}$$

Some Applications of Trigonometry — Heights and Distances

38. **38.** A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.

Answer:

$$19\sqrt{3}\text{ m}$$

Solution:

1. Height of building above boy's eyes $= 30 - 1.5 = 28.5$ m.
2. Let x_1 be the distance at 30° and x_2 be the distance at 60°.
3. Calculate $x_1 = 28.5\cot(30°)$ and $x_2 = 28.5\cot(60°)$.
4. Distance walked $= x_1 - x_2 = 28.5(\sqrt{3} - \frac{1}{\sqrt{3}}) = 28.5(\frac{2}{\sqrt{3}}) = 19\sqrt{3}$.

Derivation:

$$x_1 = 28.5\sqrt{3}, x_2 = \frac{28.5}{\sqrt{3}}$$
$$\implies \text{dist} = 28.5(\frac{3-1}{\sqrt{3}}) = \frac{57}{\sqrt{3}} = 19\sqrt{3}$$

Some Applications of Trigonometry — Two angles of elevation

39. **39.** From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. Determine the height of the tower.

Answer:

$$7(1+\sqrt{3})\text{ m}$$

Solution:

1. Let the building height $AB = 7$ m. The tower height $CD = h$.
2. From the triangle at the base, the distance between building and tower is $7\cot(45°) = 7$ m.
3. In the upper triangle, the height above the building level is $h' = 7\tan(60°) = 7\sqrt{3}$.
4. Total height $h = 7 + 7\sqrt{3}$.

Derivation:

$$h = 7 + 7\tan(60°) = 7(1+\sqrt{3})$$

Some Applications of Trigonometry — Angle of elevation and depression

40. **40.** A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60°. Find the length of the string, assuming that there is no slack in the string.

Answer:

$$40\sqrt{3}\text{ m}$$

Solution:

1. Height $h = 60$ m, angle $\theta = 60°$. Let string length be L.
2. Use sine ratio: $\sin(60°) = \frac{\text{Opposite}}{\text{Hypotenuse}} = \frac{60}{L}$.
3. Substitute $\sin(60°) = \frac{\sqrt{3}}{2}$.
4. Solve: $L = \frac{60\times2}{\sqrt{3}} = \frac{120}{\sqrt{3}} = 40\sqrt{3}$.

Derivation:

$$\begin{aligned}\sin(60°) &= \frac{60}{L}\\ \implies \frac{\sqrt{3}}{2} &= \frac{60}{L}\\ \implies L &= \frac{120}{\sqrt{3}} = 40\sqrt{3}\end{aligned}$$

Some Applications of Trigonometry — Sine ratio application